



GED 211

Risk & Return

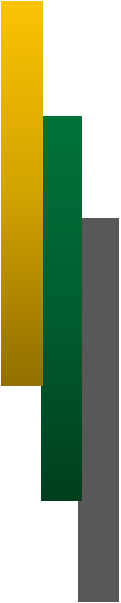
Investment & Finance

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Contents: Focus of Discussion

Risk & Return

- Concept of Risk & Return
- Types of Risk
- Probability Distribution to Measure Risk
- Attributes towards Risk
- Systematic & Unsystematic Risk
- Diversification
- CAPM
- SML
- Efficient Market



Risk & Return

Developing and Understanding

Risk & Return: Understanding Return

Return

With most investments, an individual or business spends money today with the expectation of earning even more money in the future

- Return expresses financial performance of an investment
 - The gain/loss an investor earns from an investment over a period of time

$$\text{Return} = \frac{(\text{Amount Received} - \text{Amount Invested})}{\text{Amount Invested}}$$

$$\text{Return} = \frac{(\text{Selling Price} + \text{Income} - \text{Purchase Price})}{\text{Purchase Price}}$$

This equation identifies the return over the investment holding period. In finance we commonly annualize this return

Risk & Return: Understanding Return

Example 1: You purchased a share for BDT 30 and sold it for BDT 40 a year later. Calculate your Return

$$\text{Return} = \frac{(\text{Selling Price} + \text{Income} - \text{Purchase Price})}{\text{Purchase Price}}$$

$$\text{Return} = \frac{(40 + 0 - 30)}{30} = 33.33\%$$

The investment will provide 33.33% holding period return over 1 year

Risk & Return: Understanding Return

Example 2: You purchased a share for BDT 30 and sold it for BDT 40 a year later. During this time you received dividend of BDT 2. Calculate your Return

$$\text{Return} = \frac{(\text{Selling Price} + \text{Income} - \text{Purchase Price})}{\text{Purchase Price}}$$

$$\text{Return} = \frac{(40 + 2 - 30)}{30} = 40.00\%$$

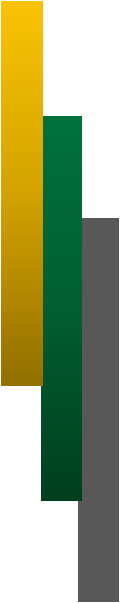
The investment will provide 40.00% holding period return over 1 year

Risk & Return: Understanding Risk

Risk

The chance that some unfavorable event will occur

- Risk is the uncertainty about the actual return that an investment will generate
 - The risk of an investor only holding one asset → Stand-Alone Risk
 - The risk of an investor holding multiple assets → Portfolio Risk
- Total Risk is measured by Standard Deviation



Risk & Return

Probability & Risk-Return

Risk & Return: Calculating Expected Return

Expected Return: Mean

DEMAND FOR THE COMPANY'S PRODUCTS	PROBABILITY OF THIS DEMAND OCCURRING	RATE OF RETURN ON STOCK IF THIS DEMAND OCCURS	
		MARTIN PRODUCTS	U.S. WATER
Strong	0.3	100%	20%
Normal	0.4	15%	15%
Weak	<u>0.3</u>	-70%	10%
	<u><u>1.0</u></u>		

$$E(Ri)_{Martin} = (0.30 * 100\%) + (0.40 * 15\%) + (0.30 * -70\%) = 15\%$$

$$E(Ri)_{US\ Water} = (0.30 * 20\%) + (0.40 * 15\%) + (0.30 * 10\%) = 15\%$$

Risk & Return: Calculating Risk

Risk: Standard Deviation

Martin Return	Expected Return	(Return – Expected Return)	Squared Differences	Probability	(Squared Difference * Probability)
100%	15%	85%	72.25%	30%	21.68%
15%	15%	0%	0%	40%	0%
-70%	15%	-85%	72.25%	30%	21.68%
Total				100%	43.35%

$$\text{Variance}_{\text{Martin}} = 43.35\%$$

$$\text{St Dev}_{\text{Martin}} = \sqrt{43.35\%} = 65.84\%$$

Risk & Return: Calculating Risk

Risk: Standard Deviation

US Water Return	Expected Return	(Return – Expected Return)	Squared Differences	Probability	(Squared Difference * Probability)
20%	15%	5%	0.25%	30%	0.075%
15%	15%	0%	0%	40%	0%
10%	15%	-5%	0.25%	30%	0.075%
Total				100%	0.15%

$$\text{Variance}_{US\ Water} = 0.15\%$$

$$\text{St Dev}_{US\ Water} = \sqrt{0.15\%} = 3.87\%$$

Risk & Return: Calculating Risk

Risk: Standard Deviation

$$St\ Dev_{Martin} = \sqrt{[(100\% - 15\%)^2 * 30\%] + [(15\% - 15\%)^2 * 40\%] + [(-70\% - 15\%)^2 * 30\%]}$$

$$St\ Dev_{Martin} = 65.84\%$$

$$St\ Dev_{US\ Water} = \sqrt{[(20\% - 15\%)^2 * 30\%] + [(15\% - 15\%)^2 * 40\%] + [(10\% - 15\%)^2 * 30\%]}$$

$$St\ Dev_{US\ Water} = 3.87\%$$

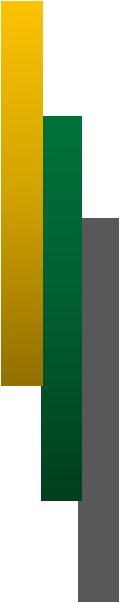
Risk & Return: Calculating Risk

Risk Per Unit of Return: Coefficient of Variation

$$CV_{Martin} = \frac{St\ Dev_{Martin}}{E(Ri)_{Martin}} = \frac{65.85\%}{15\%} = 4.39\%$$

$$CV_{US\ Water} = \frac{St\ Dev_{US\ Water}}{E(Ri)_{US\ Water}} = \frac{3.87\%}{15\%} = 0.26\%$$

While both investment has 15% expected return, US Water exhibits lower risk for the return it earns. Thus, US Water is the better investment option.



Risk & Return

Risk Aversion

Risk & Return: Attributes Towards Risk

Suppose that you must choose between the following alternatives. With alternative A, you have a 50% chance of winning \$100 and a 50% chance of losing \$100. With alternative B, you have a 50% chance of winning \$10,000 and a 50% chance of losing \$10,000

Both have an expected value equal to zero, but alternative A's outcomes have less variability

Which would you choose?

Option A: Risk Averse

Option B: Risk Loving

Indifferent: Risk Neutral

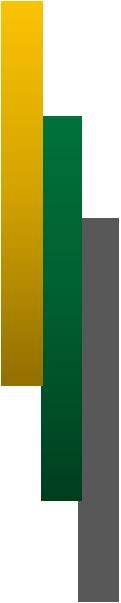
As most people are risk averse,
they would require higher return to compensate them for taking on higher risk.

Risk & Return: Attributes Towards Risk

Risk & Return Tradeoff

Higher expected returns are associated with higher levels of risk, and lower risk investments offer lower expected returns

- If you want to earn more, you must be willing to take more risk
 - Play it safe, take lower risk, expect lower return
- A higher risk has to offer higher return as compensation
 - Otherwise investors would not have the incentive to take on higher risk



Risk & Return

Types of Risk

Risk & Return: Types of Risk

While many different basis of risk typing exists, we'll be focusing on systematic & unsystematic risk, the two most relevant for the chapter

Unsystematic Risk

The risk specific to a particular company or industry and can be reduced or eliminated through diversification → Poor performance by the company, tax rate change for an industry etc.

- Also referred to as firm specific risk and diversifiable risk

Risk & Return: Types of Risk

Systematic Risk

- **Systematic Risk:** Risk that affects the entire market or economy and cannot be eliminated through diversification → interest rate change, inflation, economic downturn, exchange rate fluctuation etc.)
 - Affects all firms and industries
 - Caused by economy-wide forces
 - Cannot be diversified away
 - Measured by Beta (β)

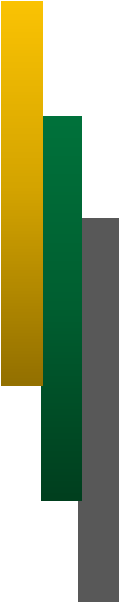
Systematic risk is also referred to as market risk and undiversifiable risk

Risk & Return: Types of Risk

Diversification

Diversification is the investment strategy of spreading funds across different assets, industries, or securities to reduce risk

- Reduces/eliminates firm and industry specific unsystematic risk
- Can not eliminate systematic risk



Risk & Return

Capital Asset Pricing Model (CAPM)

Risk & Return: (CAPM)

Capital Asset Pricing Model (CAPM)

An important tool used to analyze the relationship between risk and rates of return

- Explains the **relationship** between **expected return** and **systematic risk (beta)** of an investment
- What return an investor should expect for taking a certain level of market risk

Beta indicates how sensitive an asset is to market movements

- If the beta of a stock is 2.00, it indicates that if the market goes up by 1%, the asset will go up by 2% and if the market falls by 1% the asset will also fall by 2%
- A zero beta (extremely rare) means, the asset is insensitive to the market
- A negative beta (extremely rare) indicates inverse relationship with the market

Risk & Return: (CAPM)

CAPM - Assumptions

CAPM is based on an idealized market where the following assumptions hold:

- Investors Are **Rational** and **Risk-Averse**
- Investors Have Homogeneous (**same**) **Expectations**
 - Expected return, Variance, Correlation among investments
- **Single-Period** Investment Horizon
- **No Taxes** or Transaction Costs
- **Symmetry** of Information
- Investments are infinitely divisible
- Investors Can **Borrow** and **Lend** at the **Risk-Free Rate**
- **Unsystematic risk** can be completely eliminated through **diversification**

Risk & Return: (CAPM)

Capital Asset Pricing Model (CAPM)

$$\text{Required Return} = R_f + \beta_i(R_m - R_f)$$

$$\text{Required Return} = R_f + \beta_i * R_p$$

Example 1: If the risk free rate (R_f) is 5%, market rate of return (R_m) is 9% and the Beta (β) of a stock is 1.50, what would be the required return as per CAPM

$$\text{Required Return} = 5\% + 1.50(9\% - 5\%) = 11\%$$

Risk & Return: (CAPM)

Capital Asset Pricing Model (CAPM)

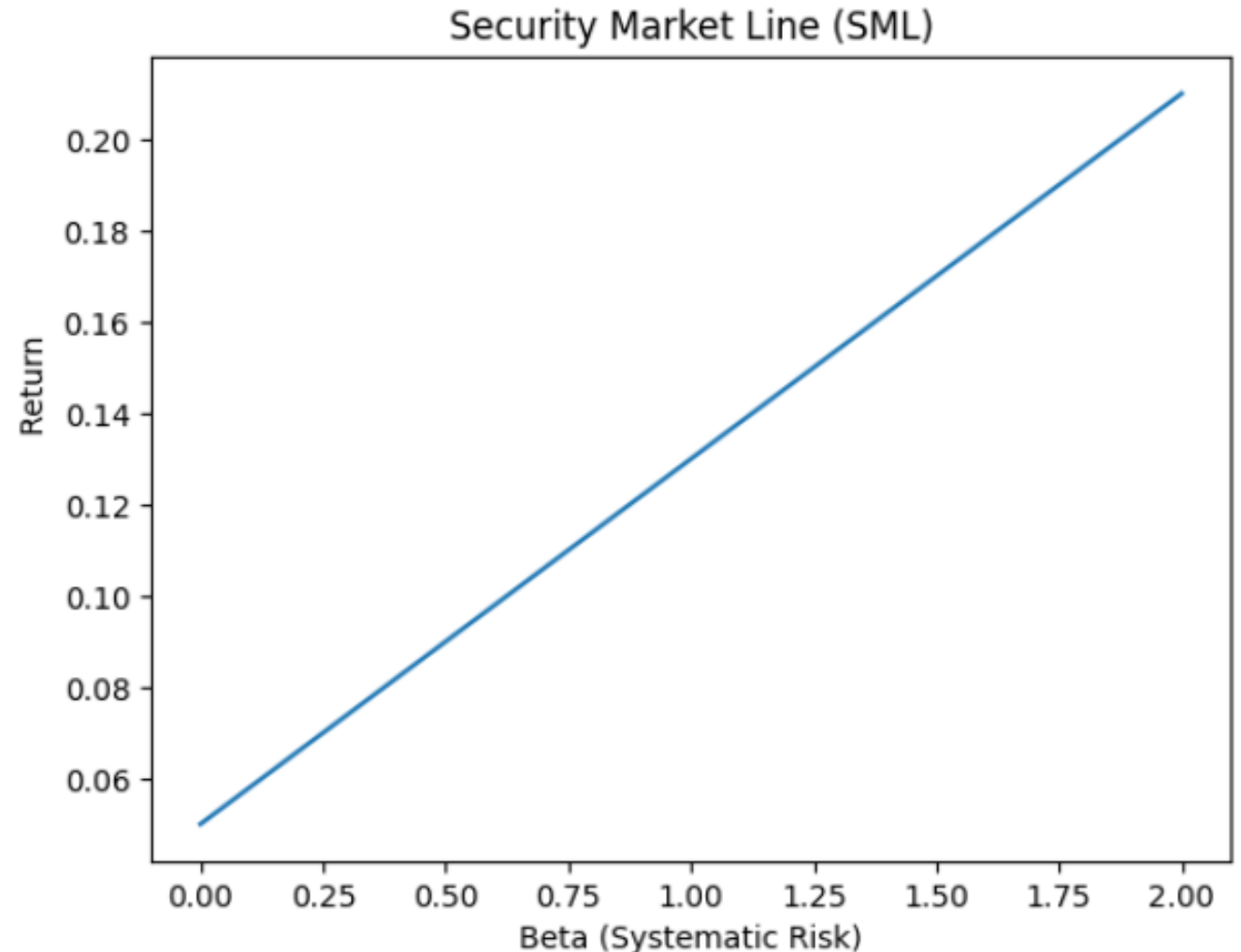
Example 1: If the risk free rate (R_f) is 5%, market risk premium (R_p) is 4% and the Beta (β) of a stock is 1.50, what would be the required return as per CAPM

$$\text{Required Return} = 5\% + 1.50 * 4\% = 11\%$$

Risk & Return: (CAPM)

Security Market Line (SML)

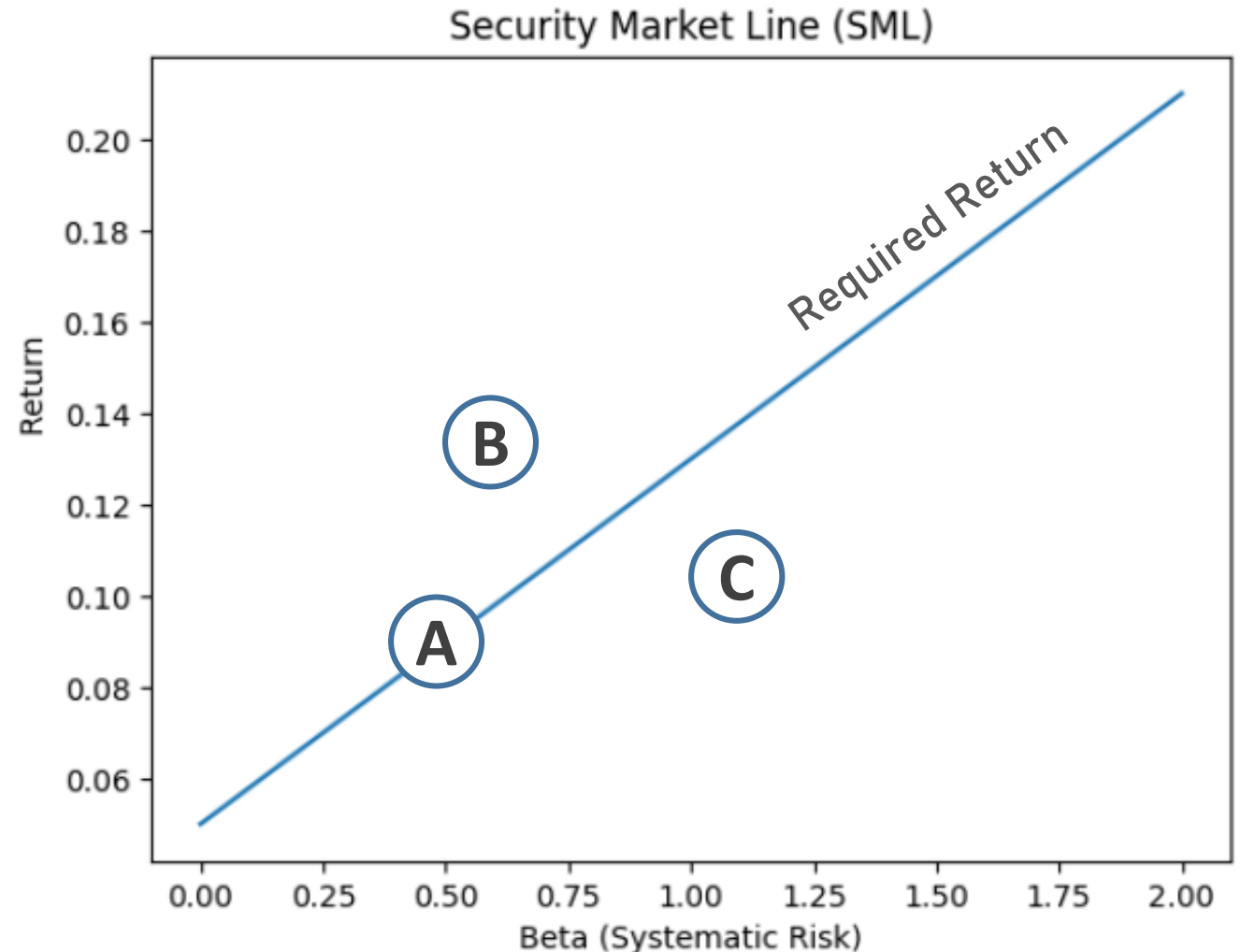
SML is a graphical representation of CAPM, that shows the relationship between expected return and systematic risk (beta)



Risk & Return: (CAPM)

Security Market Line (SML)

- Investment A: On the line
 - Expected Return = Required Return
 - Investment is correctly valued
- Investment B: Above the line
 - Expected Return > Required Return
 - Investment is undervalued
- Investment C: Under the line
 - Expected Return < Required Return
 - Investment is overvalued



Risk & Return: Efficient Market Hypothesis (EMH)

Efficient Market Hypothesis (EMH)

EMH states that financial markets fully and quickly reflect all available information in asset prices

- **Weak Form Efficiency:** Prices reflect all past market information
 - Past prices, trading volumes, price trends
- **Semi-Strong Form Efficiency:** Prices reflect all past & all publicly available information
 - Past info + Financial Statements, Public Announcements, Economic Info etc.
- **Strong Form Efficiency:** Prices reflect all information
 - Past + Public + Private Inside Information of the Companies

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